Tactix: H2 Checker-Contract Diagnostic

Report to Andrew

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Outcome

AM404 expands the fixed controls for H2 without changing the solver or its canonical proof route. A checker-only SLURM job compared the historical no-option `dsr-trim` route with `-delete-implied-units` on seven exact proof/CNF pairs. All 14 predeclared mode outcomes matched. The H2 parent-identical proof still fails no-option at the clause-60 watch-pointer error, but with the option it passes backward and forward every-lemma elaboration and independent Trestle verification. Three additional archived DSR proofs pass in both modes.

Soundness controls and provenance

The option deliberately honors deletion of an implied-unit reason. A proof whose later witnessed unit depends on that deletion is valid with the option and invalid without it. Removing only that deletion makes the same witness invalid again: Trestle reports `s INVALID` under the option. A separate five-variable bad-witness proof is likewise rejected by Trestle. These controls establish that the changed acceptance in this fixed set is caused by the specified proof-state semantics, not a blanket checker success. They are not a proof that every certificate accepted under the option is sound.

Job `1721075` ran for three seconds on hwulab with two CPUs and 8 GiB. It used the AM369 pinned checker hashes, the repository's clause-preserving checker-CNF spelling, and both backward and forward proof elaboration; successful cases required Trestle. Exact inputs, wrapper, hashes, LSRs, logs and the 14-row result table are retained under `benchmarks/probes/am404-h2-option-corpus/`. The AM404 technical report records the case matrix and its limits.

Decision and remaining gates

This makes the audited checker option a viable contract candidate, not an N2 fix. The user has been asked whether a fully audited option route may replace no-option, and whether post-finalization public clause additions should be supported or explicitly rejected. Until those choices and the associated broad proof/model checks are complete, the canonical route is unchanged. The AM400/401 full Debug suites and independent Opus affected fuzz remain active; X7 full acceptance, the source-to-action proof-rule audit, N1–N4 and the Rank/B+C/WV gates remain open. Production online learning still requires the default-exact inprocessing controller and safe-point branching/deletion strategy ownership identified earlier. The requested full Tier-D hwulab comparison at 5,000 seconds, four CPUs and 32 GiB is not submitted before goal completion.

Evidence: AM404 technical report and frozen job `1721075`; AM390 checker-contract controls; current `PROGRESS.md`; `DEVELOPMENT_PLAN.md` §5 and §8.